
New Datasets and Methods for day-in-the-life Continual Learning Benchmark

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Abstract

1 Introduction and Motivation: MARIA

Continual learning explores how AI systems can adapt and continue training and learning at deployment time. This property is necessary for systems to adapt to real-world scenarios with data distribution shifts and constantly changing information. Models operating in continual learning scenarios inherently struggle with (catastrophic) forgetting, and while there are multiple approaches for lessening or mitigating this phenomenon [1], researchers in the area have not developed a comprehensive, real-world benchmark for continual learning tasks. Most papers we are familiar with in the field operate on MNIST or CIFAR datasets and are able to prove the effectiveness of the respective methods in this very particular setup. Our goal is to test continual learning methods on valuable and interesting tasks that resemble a day in a life of a human (hence the title): finding your way around in an environment, learning new skills, reasoning through hard questions, etc. This manuscript describes our contributions along three dimensions (infrastructure, datasets, and methods) to the larger day-in-the-life benchmark. Our main additions and findings are the following:

- Sth about DER and similarity stuff
- Sth about DualPrompt
- Sth about new datasets and infrastructure for them

2 Literature Review

Continual learning methods are usually classified into five main areas – they can be based on regularization, replay (also called rehearsal), optimization, representation, and architecture [1]. This classification is not strict, and there is often overlap between the categories. To give some examples of methods from different categories, EWC [2] falls under regularization, GEM [3] is a dual replay- and optimization-based approach, Side-Tuning [4] is a representational method, and Piggyback [5] is an architectural method. The methods we explore and implement for day-in-the-life fall under replay (DER [6]) and representation (DualPrompt [7]).

2.1 Experience Replay: CHRISTIAN

2.2 Similarity Weighting: CHRISTIAN

2.3 Prompting in Continual Learning: MARIA

Prompting appeared as a representational approach instructing a pretrained model to adapt to new knowledge in the continual learning setting in L2P [8]. Instead of relying on example memorization

and replay, the knowledge about new tasks is injected into the model via training a small pool of parameters (called prompts) that are selected and added to input tokens.

DualPrompt [7] is a method inspired by L2P. It also adds a small set of new parameters (prompts) to train on top of a frozen backbone to instruct the large model to learn sequentially arriving tasks. Typically, prompts constitute 0.2-0.6% of the full model size. In DualPrompt, prompts are explicitly divided into task-specific and task-invariant instructions, which gives a sizable increase in performance as compared to the original L2P framework. All the parameters are trained together with a learnable classifier (the target task in the DualPrompt paper is classification). At test time, the prompt parameters are prepended to the embedding feature extracted from the frozen backbone model in the multi-head attention layers, which is then sent to the rest of the model.

3 Methodology

We are contributing to the `day-in-the-life` benchmark in three directions: methods that we test on the benchmark, the infrastructure of the benchmark’s repository, and datasets. The new methods were tested on the `day-in-the-life`

3.1 Backbone Model

We considered multiple base models to benchmark, fine-tune and test our CL algorithms on; these included the Qwen, SmolVLM, and Gemma model families. During our discussions, we landed on a few highest priority characteristics for selecting the default base model: ease of use and access (open-source), size (to increase the accessibility of the benchmark), quality and breadth of documentation (for dependable baseline performance), and community adoption (familiarity, edge-case debugging, guidance). Taking these criteria into account, we arrived at Qwen3-VL-2B-Instruct. This model is well documented and popular within the community, supports vision and text, and small enough to fit on the commonly-accessible Nvidia L4. In addition, its performance on many datasets lies in the “sweet spot”: not so strong that the model is saturated, obfuscating the results between different CL algorithms, and not so weak that additional training would yield negligible benefit. Finally, the suite of Qwen3-VL models range in size from 2B to 32B parameters, allowing benchmarkers with greater compute resources to upscale.

3.2 DER Implementation: CHRISTIAN

3.3 Similarity Weighting Implementation: CHRISTIAN

3.4 DualPrompt Implementation: MARIA

3.5 New Datasets

Due to the variety of datasets, GPUs, and contributors, a standard contribution and benchmarking pipeline had to be formed and agreed to. The per-dataset deliverable consists of

1. A function to download and save the raw data
2. A pre-processing function to convert the data into desired form (i.e. raw tensor -> image)
3. A loader function to load the processed dataset from a predetermined location on disk to memory
4. A configuration dictionary with context, prompting, and post-processing to ensure the LLM’s answer format matched the dataset’s provided answer format
5. Documentation on the performance of the benchmark LLM out-of-the-box, after 1 epoch of LoRA finetuning, and after 5 epochs of finetuning
6. A hook to couple the configuration dictionary to the loader function
7. A psuedo-deterministic train/eval/test split in the loader, if none is specified by the dataset authors

Following this protocol, we added eight datasets to the benchmark. Three carry native train splits and participate in the continual learning loop; the remaining five are loaded eval-only and are drawn

from the Qwen3-VL Technical Report Table 4 [9] so that base-model scores can be cross-referenced against published numbers.

The datasets we added are the following:

- **ChartQA** (Masry et al. 2022) – Visual QA over charts and plots, pre-split into `train / val / test`. The Hub version stores `label` as a plain string and `image` as `Value('binary')` bytes; preprocessing casts the image column to `datasets.Image()` and copies `label` directly to `answer`.
- **ScienceQA** (Lu et al. 2022) – Multiple-choice science QA with accompanying images. Text-only examples are filtered out and the integer `answer` index is resolved to the corresponding choice string.
- **VCR (A-OKVQA stand-in)** – The original VCR dataset (Zellers et al. 2019) requires a signed license and is not redistributable on the Hub. We substitute A-OKVQA (Schwenk et al. 2022), which has the same multiple-choice structure; the correct choice index is resolved to an `answer` string. The test split has `correct_choice_idx=None`, so `answer` is empty for test rows.
- **MMMU-Pro** (Yue et al. 2024b) – Multi-discipline, expert-level multiple-choice VQA covering subjects from art history to circuit analysis. We load the `standard` (4 options) configuration and restrict to single-image rows (the dataset can reference up to seven images per question). The stringified options list is parsed and the letter answer is resolved to its option text.
- **LogicVista** (Xiao et al. 2024) – Visual logical-reasoning multiple-choice. The options are embedded in the question text and the reference answer is a single letter, so the model is prompted to emit only A/B/C/. . . and scoring is exact-match on the letter.
- **RoboSpatial-Home** (Song et al. 2025a) – Spatial-understanding QA over indoor scenes, split into three task types: `context` (point at a region), `compatibility` (yes/no), and `configuration` (short phrase). Pointing answers are scored against a per-row binary mask, and the Qwen3-VL JSON suffix is baked into context-row questions at preprocess time so the runner stays task-agnostic.
- **ERQA** (Team et al. 2025) – Embodied-reasoning short-answer VQA, the benchmark used in the Gemini Robotics paper. We load FlagEval’s parquet mirror (the original ships as TFRecords) and restrict to single-image rows.
- **HallusionBench** (Guan et al. 2023) – Yes/no diagnostic probes for visual hallucination across charts, OCR, maps, and figures. We use the `image` split (951 rows); native 1/0 labels are mapped to yes/no so that exact-match against natural-language model replies works.

3.6 Benchmarking Protocol and Evaluation Metrics

An important part of the dataset delivery pipeline is the benchmarking step. The benefits include data format debugging, validation of the Qwen3 published benchmark score, a mental heuristic for the dataset questions, and a comparison point for the CL algorithm vs a simple LoRA SFT approach.

The first step is matching the experiment setup to the conditions used by the Qwen researchers when conducting the baseline evaluation. In the case that we created our own train/eval/test splits, we concatenate the splits; the Qwen3 benchmarking was done **across the entire dataset**. Additional sources of noise include thinking vs instruct mode, the maximum internal sequence length, and the scoring function (exact match accuracy, relaxed match accuracy, MSE, CE loss, etc)

The second step is drafting an eval config that results in optimal baseline performance. For example, most datasets expect a single-token response. In the absence of additional instruction, the LLM tends to produce a verbose answer, which fails the regex match every time. Providing clear instruction denoises the training signal in backprop, improving performance and interpretability.

The third step is to iteratively run the baseline evaluation until the score matches the published Qwen3 number. This is a sanity check for the dataset and the evaluation pipeline. **Many** bugs were caught and fixed during this step.

The fourth step is to run a simple LoRA SFT on the dataset for one and five epochs. This step places the model on the underperformance <=> saturation spectrum.

The final step is to record the results in the `day_in_the_life/datasets/README.md` table. In addition to the quantitative results included in the table, qualitative observations, shortcuts, compromises, caveats, etc. should be appended to the bottom section

Benchmark Performance								
Dataset	Metric	Model	Approach	Hardware	Score (No-Finetune)	Score (Finetune 1 epoch)	Score (Finetune 5 epochs)	Notes
ChartQA	Accuracy	Qwen/Qwen3-VL-4B-Instruct	QLoRA (4-bit base, LoRA $r=16$); 2K train / 500 test subsample; vLLM eval	1x NVIDIA L4 (24 GB VRAM), 125 GB RAM, 32 vCPU	0.7240	0.7100	0.7120	Not in Qwen3-VL Tech Report Table 4 — see Methodology investigation
ERQA	Accuracy	Qwen/Qwen3-VL-4B-Instruct	QLoRA (4-bit base, LoRA $r=16$); 2K train / 500 test subsample; vLLM eval	1x NVIDIA L4 (24 GB VRAM), 125 GB RAM, 32 vCPU	0.4390	0.4667	0.4667	Paper: 0.413 · ours: 0.439

3.7 Changes to Processing Existing Datasets: Maria

4 Experimental Results

4.1 Qwen-4b Performance Reproduction

Before testing any CL algorithms, it's necessary to ensure that the baseline performance of the Qwen3-VL-2B-Instruct model on the datasets matches the published numbers.

4.2 New Methods' Performance: CHIRSTIAN and MARIA

4.3 Ablations for Replay and Similarity Weighting: CHIRSTIAN

4.4 Learning Rate Ablations for Baseline Training: MARIA

4.5 DualPrompt Ablations: MARIA

5 Further Discussion

5.1 Challenges and Limitations: CHRISTIAN

5.2 Future Directions: AKSEL

6 Conclusion

6.1 Summary of Findings: ALL

6.2 Contributions: ALL

Contributions of each team member:

- Aksel: Initial integration tests, cleanup, and bug fixes; contribution and evaluation of 8 novel datasets; introduction and implementation of standard dataset onboarding pipeline; scoping and selection of default VLM model. *Future Work* pytorch-lightning training loop implementation; model-merging CL algorithm, following the example from “A Practitioner’s Guide to Continual Multimodal Pretraining.” [10]
- Christian:
- Maria:

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Code Availability

All the code used in this project is available at <https://github.com/AmoghSInamdar/day-in-the-life>.

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Appendix